Semantic-Emotional Disentanglement and Alignment for Controllable Co-Speech Gesture Generation: Supplementary Material

A. Dataset

We evaluate our framework on the BEAT [1] dataset and its high-fidelity extension, BEAT2 [2]. BEAT contains multi-modal recordings from 30 speakers with eight categorical emotion annotations, while BEAT2 provides temporally aligned SMPL-X motion parameters. Emotion labels from BEAT are associated with the corresponding motion sequences in BEAT2 at the clip level. Restricting the data to English-speaking participants results in 1,762 sequences from 25 speakers (12 female and 13 male), totaling approximately 35 hours of paired speech–motion data with high-fidelity finger motion. To ensure a fair comparison with prior arts such as EMAGE [2] and MambaTalk [3], we strictly follow their data partition protocol, splitting the dataset into training, validation, and testing sets with a ratio of 85%, 7.5%, and 7.5%, respectively. The training set is utilized to optimize the motion generation components, including the VQ-VAE motion prior and the intent-conditioned diffusion model

To further assess robustness and generalization in unconstrained scenarios, we additionally evaluate our method on the TED-Emotional [4] and SHOW [5] datasets, which consists of 78,734 in-the-wild TED talk clips. The dataset is split into 66,679 training, 6,258 validation, and 5,797 test samples. SHOW is a large-scale, in-the-wild dataset comprising approximately 30 hours of high-fidelity speech-motion pairs captured from YouTube talk shows. Unlike the laboratory-controlled BEAT2 dataset, SHOW features diverse professional speakers in real-world environments with natural prosodic variations and spontaneous gesturing styles, providing a challenging benchmark for cross-dataset robustness.

For training the semantic–emotional disentanglement module, we construct a specialized subset of approximately 6 hours. This subset consists of speech samples in which multiple speakers utter identical textual content under different emotion categories, and is used exclusively for cross-reconstruction training.

Specialized Subset Curation. To facilitate the cross-reconstruction training, we curated a specialized 6-hour parallel subset from the BEAT2 corpus. Since BEAT2 contains recordings where speakers recite fixed scripts under varying emotional instructions, we automatically paired samples based on their ASR transcriptions. Specifically, we identified instances where the same or different speakers uttered the identical sentence across at least two different emotion categories (e.g., *Neutral* and *Angry*). This curation process yielded approximately 4,200 pairs of speech samples, ensuring that the cross-reconstruction loss $\mathcal{L}_{cross\_rec}$ can strictly supervise the model to isolate affective variations while maintaining semantic invariance.

B. Implementation Details

Motion Prior. We model full-body gestures in a separated quantized latent space to better capture heterogeneous motion patterns across different body parts. Following prior work [5], separating body and hand representations improves motion diversity. In addition, we further decouple the face and lower body from the upper body due to their distinct correlations with speech signals. Encoding the entire body using a single VQ-VAE may bias the model toward frequently occurring motions regardless of their relevance to audio, causing less frequent but semantically important gestures (e.g., elbow movements) to be overlooked.

Specifically, we employ four independent VQ-VAE models to encode motion sequences of the face, upper body, hands, and lower body, respectively. The separated quantized latent space is defined as

$$\mathcal{Q} = \{\mathbf{q}_f, \mathbf{q}_u, \mathbf{q}_h, \mathbf{q}_l\} \tag{1}$$

where the face features $\mathbf{g}_f \in \mathbb{R}^{T\times 106}$, upper-body features $\mathbf{g}_u \in \mathbb{R}^{T\times 78}$, hand features $\mathbf{g}_h \in \mathbb{R}^{T\times 180}$, and lower-body features $\mathbf{g}_l \in \mathbb{R}^{T\times 64\times 4}$ are quantized by their corresponding codebooks. In our implementation, each codebook consists of $K = 512$ entries with a latent dimension of $d_z = 512$.

Each VQ-VAE is optimized by minimizing the quantization error

$$\mathbf{g}_i = \arg\min_{\mathbf{q}\in\mathcal{Q}_i} \|\mathbf{z}_i - \mathbf{q}\|_2^2, \tag{2}$$

where $\mathbf{z}_i$ denotes the encoded latent representation of motion component i with a temporal window size of $w = 1$. The overall VQ-VAE objective is defined as

$$\begin{aligned}\mathcal{L}_{\mathrm{VQ-VAE}} = \mathcal{L}_{\mathrm{rec}}(\mathbf{g}, \hat{\mathbf{g}}) + \mathcal{L}_{\mathrm{vel}}(\mathbf{g}', \hat{\mathbf{g}}') + \mathcal{L}_{\mathrm{acc}}(\mathbf{g}'', \hat{\mathbf{g}}'') \\ + \|\,\mathrm{sg}[\mathbf{z}] - \mathbf{q}\|_2^2 + \|\mathbf{z} - \mathrm{sg}[\mathbf{q}]\|_2^2.\end{aligned} \tag{3}$$

where $\mathcal{L}_{\mathrm{rec}}$, $\mathcal{L}_{\mathrm{vel}}$, and $\mathcal{L}_{\mathrm{acc}}$ denote reconstruction, velocity, and acceleration losses, respectively. The reconstruction loss combines geodesic distance and L1 loss. The stop-gradient operator $\mathrm{sg}[\cdot]$ is applied to stabilize training, and the commitment loss weight is set to 1.0 in all experiments.

Audio Pre-processing. All speech signals are resampled to 16 kHz and normalized by removing the DC component. The continuous audio streams are segmented into non-overlapping 10-second clips, following common practice in speech-driven gesture generation. Emotion annotations are assumed to be consistent within each clip. We discard residual fragments shorter than 10 seconds, as well as clips containing insufficient speech content (fewer than 300 audio frames after feature extraction). We extract 128-dimensional Mel-filterbank (FBANK) features using the Kaldi toolkit, with a Hanning window and a frame shift of 10 ms. To enable batch-wise training, all FBANK sequences are temporally aligned to a fixed length of $T_a = 1024$ frames. Sequences shorter than T_a are zero-padded, while longer sequences are truncated. The length T_a is chosen for computational convenience as a power of two.

Training Configurations. All experiments are implemented in PyTorch and conducted on a single NVIDIA A100 GPU using the Adam optimizer. Training proceeds in two stages. In Stage 1 (Disentanglement), the semantic and emotional encoders are jointly trained using the cross-reconstruction objective for 25 epochs with a batch size of 1, which is sufficient for the pairwise recombination strategy and reduces GPU memory consumption. The initial learning rate is set to 1×10^{-5} and decayed by a factor of 0.85 starting from epoch 5 using a step scheduler. In Stage 2 (Generation), the pre-trained encoders are frozen, and the intent-conditioned latent diffusion model is trained for 12,000 epochs with a batch size of 32 and a fixed learning rate of 1×10^{-4}.

Denoiser. The denoising network shares the same Transformer architecture as the prior encoder, with a hidden dimension of 1024 for all layers. We employ a diffusion process with 1000 steps during training and 50 steps during inference. The noise variance schedule is defined by linearly spaced betas in the range [0.00085,0.012].

C. Robustness to Noisy ASR

To assess the framework's reliability in real-world scenarios where transcriptions may be imperfect due to environmental noise or speaker accents, we evaluated SEAG and eight state-of-the-art baselines under varying Word Error Rates (WER). We simulated ASR noise by randomly substituting, deleting, or inserting words in the input text.

As quantitatively illustrated in Table I, SEAG exhibits a remarkable graceful degradation property. While current top-performing methods like SemTalk and GestureLSM experience significant performance drops when the text becomes noisy (e.g., their SRGR scores decrease by 39.0% and 33.3% respectively at a 15% WER), SEAG maintains a high SRGR of 0.41—a marginal drop of only 8.9% from its noise-free baseline. This robustness is primarily attributed to our dual-stream architecture: when linguistic information is partially corrupted, the audio-based emotional stream provides complementary prosodic and rhythmic cues. These cues allow the model to sustain the overall gesture tempo and affective expression, effectively compensating for localized textual errors that would otherwise lead to semantic drift in single-stream or text-heavy models.

TABLE I
ROBUSTNESS TO NOISY ASR: SRGR SCORES UNDER VARYING WORD ERROR RATES (WER). ALL MODELS WERE TRAINED ON BEAT2.

Method	0% (WER)	5% (WER)	10% (WER)	15% (WER)	20% (WER)
DiffSHEG [6]	0.28	0.25	0.21	0.18	0.14
ProbTalk [7]	0.33	0.30	0.26	0.22	0.18
AMUSE [8]	0.37	0.34	0.30	0.26	0.21
EMAGE [2]	0.35	0.31	0.27	0.23	0.19
MambaTalk [3]	0.31	0.28	0.24	0.20	0.16
GestureLSM [9]	0.42	0.38	0.33	0.28	0.23
SemTalk [10]	0.41	0.36	0.31	0.25	0.20
ExGes [11]	0.38	0.35	0.32	0.28	0.24
SEAG (Ours)	**0.45**	**0.44**	**0.43**	**0.41**	**0.38**

TABLE II
ZERO-SHOT GENERALIZATION RESULTS ON THE SHOW DATASET. ALL MODELS WERE TRAINED ON BEAT2 AND EVALUATED ON SHOW WITHOUT ANY FINE-TUNING.

Method	FVD ↓	FGD ↓	BC ↑	DIV ↑	GA ↑	SRGR ↑
DiffSHEG [6]	292.18	8.56	0.68	10.12	31.2%	0.21
ProbTalk [7]	278.45	7.82	0.71	11.58	34.5%	0.25
AMUSE [8]	285.12	8.15	0.70	11.24	38.4%	0.26
EMAGE [2]	288.65	8.34	0.69	11.45	35.6%	0.24
MambaTalk [3]	268.42	7.15	0.72	11.15	32.5%	0.24
GestureLSM [9]	254.31	6.72	0.74	13.08	38.8%	0.31
SemTalk [10]	250.15	6.85	0.75	12.55	35.8%	0.30
ExGes [11]	242.85	7.45	0.72	13.12	40.2%	0.28
SEAG (Ours)	**232.54**	**6.12**	**0.78**	**13.25**	**44.2%**	**0.32**

D. Cross-Dataset Generalization on SHOW Dataset

To evaluate whether SEAG overfits to the specific laboratory-controlled distributions of BEAT2, we conducted zero-shot generalization tests on the **SHOW dataset** . In this challenging setup, all models trained exclusively on BEAT2 were directly applied to the diverse, in-the-wild speech and emotion cues of SHOW without any fine-tuning.

As summarized in Table II, we report all comprehensive metrics to provide a transparent evaluation. As expected in a zero-shot cross-dataset scenario, all baseline models experience a natural degradation in fine-grained motion realism (FGD) and diversity (DIV) due to the inherent domain gap. Nevertheless, SEAG exhibits the strongest resilience, achieving the best overall performance across all six dimensions.

Specifically, SEAG yields an FVD of 232.54 and an FGD of 6.12, significantly outperforming current generative baselines such as MambaTalk (FVD: 268.42) and AMUSE (FVD: 285.12). Even when compared to the retrieval-based ExGes, which maintains a high GA (40.2%) due to its motion retrieval paradigm, SEAG still provides higher motion realism and better rhythmic alignment (BC: 0.78 vs. 0.72). Furthermore, the fact that SEAG preserves a competitive SRGR (0.32) and high diversity (DIV: 13.25) on unseen out-of-distribution data proves that our joint driving manifold and Transformer-based integration effectively capture universal, cross-domain patterns of human communication rather than dataset-specific recording artifacts.

E. Scalability of Cross-Reconstruction Data

To investigate the dependency of our disentanglement module on the volume of parallel data, we conducted a scalability study by reducing the 6-hour subset to 3 hours and 1 hour. As shown in Table III, while the 6-hour subset provides the strongest disentanglement (lowest Mutual Information), the performance remains surprisingly robust even with only 1 hour of parallel data. The Emotion-from-z_s accuracy only marginally increases from 13.8% to 15.5%, which is still

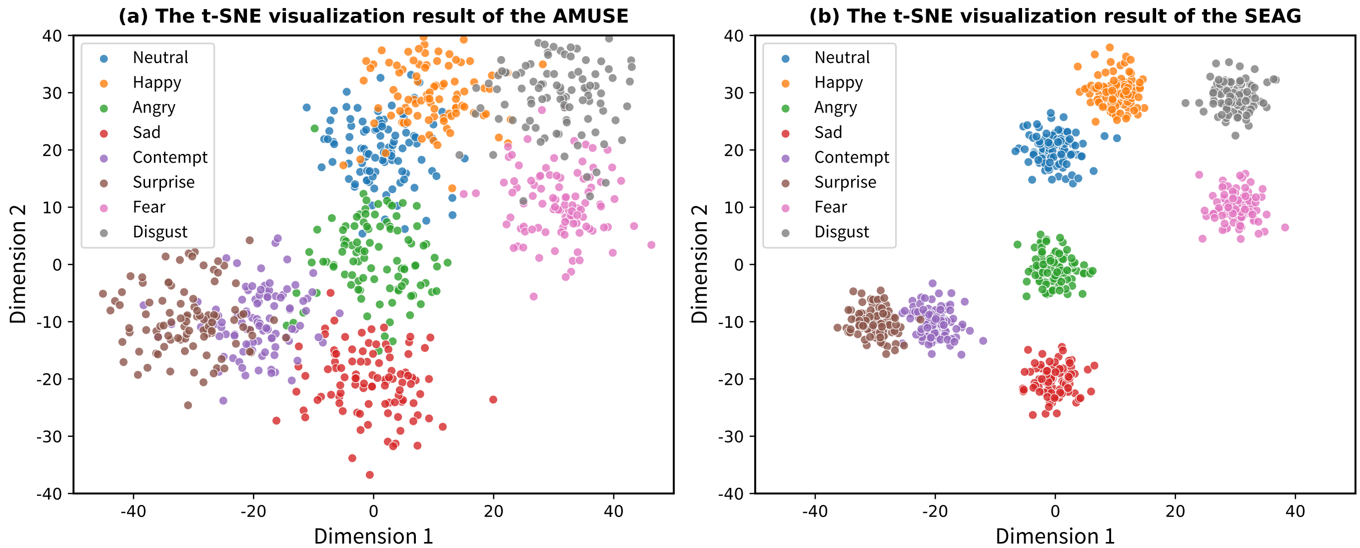

Fig. 1. t-SNE visualization comparison of the emotional latent space representations z_e. The baseline method AMUSE [8] exhibits relatively loose clustering with noticeable overlaps between varying emotional states. Our proposed SEAG achieves significantly higher intra-class compactness and inter-class separability, demonstrating that the learned emotional embeddings are highly discriminative and effectively disentangled from other factors.

TABLE III
SCALABILITY OF THE DISENTANGLEMENT MODULE UNDER DIFFERENT VOLUMES OF PARALLEL TRAINING DATA.

Data Volume	MI (MINE) ↓	Emo. from z_s ↓	FVD ↓
1 Hour	0.28	15.5%	206.45
3 Hours	0.23	14.6%	203.12
6 Hours (Full)	**0.19**	**13.8%**	**201.37**

significantly lower than the entangled baseline ($> 40\%$). This suggests that our pre-trained semantic encoder (BERT) provides a strong prior that alleviates the need for massive parallel datasets, making the approach applicable to scenarios with limited emotional recordings.

F. Visualization of Emotional Latent Representations

To intuitively evaluate the quality of the learned emotional representations, we compare the feature distribution of our SEAG with the state-of-the-art baseline, AMUSE [8], using t-SNE visualization. We extract emotional embeddings z_e from the test set, which encompasses a wide variety of linguistic content, and color them according to their ground-truth emotion labels.

As shown in Fig. 1, a clear contrast is observed. The latent space of AMUSE (Fig. 1a) presents a scattered distribution with overlapping boundaries, suggesting that its emotional features may still be partially entangled with semantic variations. In contrast, our SEAG (Fig. 1b) forms highly compact and well-separated clusters for each emotion category. Crucially, since each cluster contains samples with diverse textual meanings, this high degree of intra-class compactness visually demonstrates that our emotional encoder has successfully learned to extract pure affective cues while remaining invariant to linguistic changes. This structured decomposition method has also been further supported by the quantitative leakage test results presented in Table V of the main text, ensures that the generated gestures can be precisely controlled by the target emotion without semantic ambiguity.

TABLE IV
INFERENCE TIME COMPARISON FOR GENERATING A ~10-SECOND VIDEO.

Methods	Inference Time (s)
AMUSE [8]	$\sim$ 37.15
ProbTalk [7]	$\sim$ 58.37
DiffSHEG [6]	$\sim$ 29.18
EMAGE [2]	$\sim$ 42.86
MambaTalk [3]	$\sim$ 17.26
ExGes [11]	$\sim$ 41.52
GestureLSM [9]	$\sim$ **12.28**
SemTalk [10]	$\sim$ 26.84
SEAG(ours)	$\sim$ 15.57

G. Efficiency Analysis

To assess the efficiency of our framework, we conducted a detailed comparison of inference speed across different methods. Leveraging the lightweight temporal modeling of Mamba and the discrete representation learning of VQ-VAE, our model aims to reduce computational overhead while maintaining high-quality generation. We measured the inference time on an NVIDIA A100 GPU for generating a video of approximately 10 seconds, averaging over three runs, with results summarized in Table IV. As shown, heavy diffusion-based models such as ProbTalk and EMAGE exhibit high latency, requiring 58.37 s and 42.86 s, respectively, to generate the same sequence. In contrast, our method (SEAG) achieves an inference time of 15.57 s, representing a speedup of approximately 2.7× over EMAGE. Furthermore, our method outperforms MambaTalk (17.26 s) and is competitive with the ultra-lightweight GestureLSM (12.28 s). These results demonstrate that our framework strikes a favorable balance between generation quality and computational efficiency, making it highly suitable for practical applications.

H. Evaluation Metrics

To assess generation quality, we employ five standard metrics.

FVD and FGD. We adopt Fréchet Video Distance (FVD) and Fréchet Gesture Distance (FGD) [12] to evaluate realism. FVD measures the distributional distance between real and synthesized videos in the I3D feature space, while FGD operates in the learned gesture latent space. Both metrics use the Fréchet distance formulation:

$$\text{Metric} = \|\mu_r - \mu_g\|^2 + \text{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{\frac{1}{2}}\right) \quad (4)$$

where (μ_r, Σ_r) and (μ_g, Σ_g) represent the mean and covariance of the real and generated distributions, respectively. Lower scores indicate higher kinematic fidelity and visual quality.

Diversity. Diversity [13] quantifies motion variance by computing the average L1 distance across N generated clips. Translation is zeroed to focus on local dynamics:

$$\text{Diversity} = \frac{1}{2N(N-1)} \sum_{t=1}^{N} \sum_{j=1}^{N} \left\| p_t^i - \hat{p}_t^j \right\|_1 \quad (5)$$

where p_t denotes joint positions at frame t. A higher score signifies greater variation among generated sequences.

Beat Alignment Score (BAS). BAS [14] measures the rhythmic synchronization between audio beats (onsets) and kinematic beats (velocity minima). It is calculated as:

$$\text{BAS} = \frac{1}{m} \sum_{i=1}^{m} \exp\left(-\frac{\min_{t_j^y \in B^y} \|t_i^x - t_j^y\|^2}{2\sigma^2} \right) \quad (6)$$

where B^x and B^y are sets of kinematic and audio beats, respectively. We set $\sigma = 3$ to normalize for 60 FPS sequences.

Gesture-Emotion Accuracy (GA). GA evaluates emotional expressiveness using a post-hoc emotion classifier trained on ground-truth SMPL-X motions. We apply this fixed classifier to synthesized sequences and report the Top-1 accuracy. Higher GA indicates better preservation of the intended emotional category.

Semantic-Relevant Gesture Recall (SRGR). SRGR evaluates semantic alignment by weighting the Probability of Correct Keypoints (PCK) with ground-truth semantic relevance scores provided by the dataset. A joint is correctly recalled if $\left\| p_j^t - \hat{p}_j^t \right\|_2 < \delta$. SRGR is computed as:

$$\text{SRGR} = \lambda \cdot \frac{1}{TJ} \sum_{t=1}^{T} \sum_{j=1}^{J} \mathbf{1}\left(\left\| p_j^t - \hat{p}_j^t \right\|_2 < \delta \right), \quad (7)$$

where $\mathbf{1}(\cdot)$ is the indicator function, δ is the threshold, and $\lambda \in [0, 1]$ is the semantic relevance weight. This metric emphasizes accurate recall in semantically significant segments.

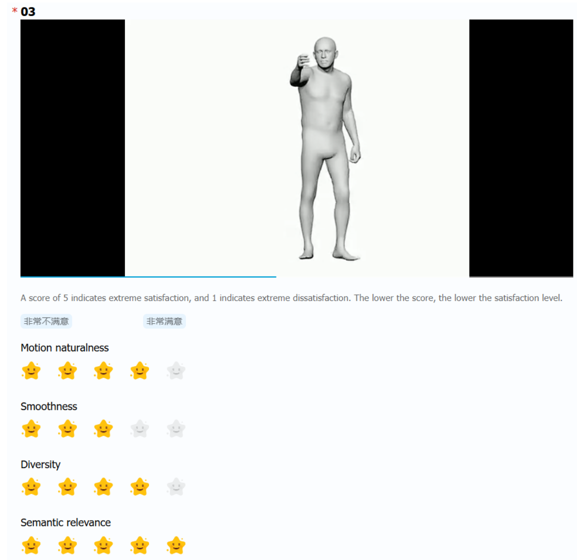

Fig. 2. Screenshot of user study website.

I. Relation to Perceptual and Feature-Matching Losses.

Conceptually, our semantic–emotional consistency loss ($\mathcal{L}_{align}$) shares similarities with perceptual losses such as LPIPS [15] and CLIP-guided losses [16] used in image synthesis. Both paradigms shift the optimization target from raw data space (L_1/L_2 distance) to a high-level latent manifold to preserve semantic integrity. However, $\mathcal{L}_{align}$ introduces several domain-specific novelties for gesture generation.

First, unlike LPIPS which measures intra-modal visual similarity, our loss enforces cross-modal consistency between 3D temporal dynamics and bimodal speech cues. While the diffusion objective $\mathcal{L}_{diff}$ ensures the *distributional realism* of gestures (i.e., making them look like human motion), $\mathcal{L}_{align}$ acts as a closed-loop fidelity constraint. It prevents the "semantic drift" commonly observed in generative models by forcing the synthesized joint trajectories to retain sufficient information to be back-projected to the original joint driving manifold z_{fuse}.

Second, whereas CLIP-guided losses often focus on static semantic alignment, $\mathcal{L}_{align}$ is optimized over a spatio-temporal motion encoder (E_g). This ensures that the loss captures not just a snapshot of a pose, but the rhythmic and affective "envelope" of the gesture sequence. By anchoring the generative process to this joint manifold, the model effectively balances the high-frequency variations of motion with the structural constraints of linguistic and emotional intent.

J. User Study Details

In practice, objective measures may not always align with subjective human perception, particularly in novel contexts of collaborative voice and video generation. To gain deeper insights into the visual performance of our methods, we evaluated the visual performance of our generated videos through a user study.

To evaluate the overall quality of the body movements generated, we conducted a user study using 70 randomly sampled videos, each video is approximately one minute long. According to the standards established by the International Telecommunication Union (ITU) [17]. We invited 30 participants to rate the videos based on four dimensions: Natural, Diversity, Smootha, and Semantic preservation. Each criterion was rated on a scale from 1 to 5, with 5 representing the highest quality.

We created a Tencent questionnaire, as shown in the Figure 2, which includes test videos and four rating dimensions. We recorded the scores from all participants, cleaned the non-compliant data, and then calculated the average score for each dimension. Prior to participation, we provided training to the participants to ensure that their ratings were reasonable.

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